

18.4.4 Alternate Definition of Expectation

There is another standard way to define expectation.

Theorem 18.4.3. *For any random variable R ,*

$$\text{Ex}[R] = \sum_{x \in \text{range}(R)} x \cdot \Pr[R = x]. \quad (18.3)$$

The proof of Theorem 18.4.3, like many of the elementary proofs about expectation in this chapter, follows by regrouping of terms in equation (18.2):

Proof. Suppose R is defined on a sample space $\mathcal{S}$. Then,

$$\begin{aligned} \text{Ex}[R] &::= \sum_{\omega \in \mathcal{S}} R(\omega) \Pr[\omega] \\ &= \sum_{x \in \text{range}(R)} \sum_{\omega \in [R=x]} R(\omega) \Pr[\omega] \\ &= \sum_{x \in \text{range}(R)} \sum_{\omega \in [R=x]} x \Pr[\omega] && (\text{def of the event } [R = x]) \\ &= \sum_{x \in \text{range}(R)} x \left(\sum_{\omega \in [R=x]} \Pr[\omega] \right) && (\text{factoring } x \text{ from the inner sum}) \\ &= \sum_{x \in \text{range}(R)} x \cdot \Pr[R = x]. && (\text{def of } \Pr[R = x]) \end{aligned}$$

The first equality follows because the events $[R = x]$ for $x \in \text{range}(R)$ partition the sample space $\mathcal{S}$, so summing over the outcomes in $[R = x]$ for $x \in \text{range}(R)$ is the same as summing over $\mathcal{S}$. ■

In general, equation (18.3) is more useful than the defining equation (18.2) for calculating expected values. It also has the advantage that it does not depend on the sample space, but only on the density function of the random variable. On the other hand, summing over all outcomes as in equation (18.2) sometimes yields easier proofs about general properties of expectation.

18.4.5 Conditional Expectation

Just like event probabilities, expectations can be conditioned on some event. Given a random variable R , the expected value of R conditioned on an event A is the probability-weighted average value of R over outcomes in A . More formally: